

CSE472: Machine Learning

Date: 9 December 2023

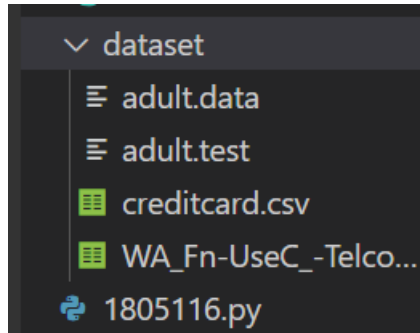
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Instructions to train and test the models:

1. Please run **1805116.py** to run all experiments and get required metrics as mentioned in specifications. It will print the metrics of the datasets first trained and tested using only logistic regression and AdaBoost sequentially. It may take some time.

```
546 # -----Data Preprocessing-----
547
548 # -----Telco Churn Dataset-----
549 train, test, output_col = telco_preprocessing()
550 # -----Adult Dataset-----
551 # train, test, output_col = adult_preprocessing()
552 # -----Credit Card Dataset-----
553 # train, test, output_col = creditcard_preprocessing()
554 # -----Logistic Regression-----
555
556 epochs = 5000
557 lr = 0.5
558 threshold = 0
559 # best feature count for ds_telco=10, ds_adult=5, ds_credit=8
560 best_features_count = 5
561 # w, top_k_features, acc = logistic_regression(train, test, epo
562 w, top_k_features = train_model(train, output_col, epochs, lr,
563
602 # # -----Adaboosting-----
603 # hypothesis_weight, hypothesis_inernal_weights, hypothes
604 # k_iter_list=[5,10,15,20]
605 # for k in k_iter_list:
606 #     print(".....",k,".....")
607 #     hypothesis_weight_k=hypothesis_weight[:k]
608 #     hypothesis_inernal_weights_k=hypothesis_inernal_weigh
609 #     hypothesis_top_k_features_k=hypothesis_top_k_featur
610 #     y_predicted, acc = adaboosting_test(test, hypothesis
611 #     y_pred, acc1 = adaboosting_test(train, hypothesis_w
612 #     print("Accuracy in train:", acc1)
613 #     print("Accuracy in test:", acc)
614
615
```

2. If it is required to run on a specific dataset with specific hyperparameters, some functions are commented out in the **test()** function [Line 603-614]. Please uncomment according to whichever dataset is necessary and whichever model is necessary to run any of the models (LR, AdaBoost) on that specific dataset and view performance metrics.
3. Please uncomment only the required function calls for decreasing runtime.
4. Please keep the folder of dataset files named “dataset” in the same directory as the 1805116.py. The file names should be the original file names.



Performance measure of logistic regression:

Feature Selection = 10/5/8, Epochs = 5000, Learning Rate = 0.5 (Constant), No early stopping

Telco Churn Dataset:

Performance Measure	Training	Test
Accuracy	0.7635782747603834	0.7324343506032647
Recall	0.31815091774303195	0.27889447236180903
Specificity	0.9209704539995196	0.9109792284866469
Precision	0.5872020075282308	0.5522388059701493
False Discovery Rate	0.4127979924717692	0.4477611940298507
F1 score	0.4126984126984127	0.37061769616026713

Credit Card Fraud Dataset:

Performance Measure	Training	Test
Accuracy	0.9956355433503334	0.9992951045007578
Recall	0.7933673469387755	0.7777777777777778
Specificity	0.9996	0.9996117532868614
Precision	0.9749216300940439	0.7411764705882353
False Discovery Rate	0.025078369905956133	0.2588235294117647
F1 score	0.8748241912798874	0.7590361445783134

Adult Census Dataset:

Performance Measure	Training	Test
Accuracy	0.7477333497249286	0.7572648522454998
Recall	0.744355147340222	0.7527301092043682
Specificity	0.74880557130132	0.7586678465127504
Precision	0.48467480687764763	0.4910941475826972
False Discovery Rate	0.5153251931223524	0.5089058524173028
F1 score	0.5870811952912767	0.5943948259932246

Performance measure of AdaBoost implementation:

Feature Selection =6, Epochs = 2, Learning Rate = 0.0001 (Constant),
Early Stopping Threshold = 0.5

Telco Churn Dataset:

Number of boosting rounds	Training	Test
5	0.7389066382676607	0.7175301632363378
10	0.7389066382676607	0.7175301632363378
15	0.7389066382676607	0.7175301632363378
20	0.7389066382676607	0.7175301632363378

Credit Card Fraud Dataset:

Number of boosting rounds	Training	Test
5	0.8413103177716752	0.843090261868678
10	0.9109454688112986	0.9121876431818983
15	0.9545409964692037	0.9595389983434955
20	0.9694488034523342	0.9749409650019385

Adult Census Dataset:

Number of boosting rounds	Training	Test
5	0.7590742846605403	0.7637156724212079
10	0.7590742846605403	0.7637156724212079
15	0.7590742846605403	0.7637156724212079
20	0.7590742846605403	0.7637156724212079

Observations:

1. Logistic regression accuracy increases with decrease of number of best features for “adult” and “credit card” dataset.
2. The AdaBoost accuracy increases with the increase of the boosting rounds in one(credit card) dataset, the remaining two doesn't get affected with increasing boosting rounds.
3. The negative bias of the datasets also impacts the specificity of the models, as can be seen in the logistic regression analysis.